

PROJECT REQUIREMENTS DOCUMENT

Project Name	Ecosystem Discovery Application
---------------------	---------------------------------

Functional Requirements	
The system shall display a web-based interactive freshwater ecosystem scene representing a single Saskatchewan-inspired freshwater environment, including sky, shoreline, water surface, and underwater zones.	
The system shall display 6–8 clickable animal sprites placed within ecologically appropriate zones of the ecosystem scene.	
The system shall present an animal information card (pop-up panel) when a user clicks an animal, displaying: • Animal name • Cartoon-style image • Short, age-appropriate description (reading level 7–9 years)	
The system shall provide a sticker-book collection interface that displays discovered animals as visible stickers and undiscovered animals as locked or hidden placeholders.	
The system shall display the user’s overall discovery progress (e.g., “3 of 6 animals discovered”).	
The system shall allow users to navigate between the ecosystem scene and the sticker-book panel within the same interface layout.	
The system shall provide navigation allowing users to switch between the ecosystem scene and the sticker-book collection page.	
The system shall use child-friendly design elements including large clickable areas, bright visuals, and simple text suitable for ages 5–9.	
The system shall use child-friendly design principles including: • Large clickable areas • Bright but readable visuals • Simple navigation • Clear interaction feedback	
The system shall persist discovery progress during a user session and restore progress when the page is revisited (using structured data storage).	
The system shall support the complete discovery flow: Ecosystem exploration → Animal selection → Information display → Sticker-book update → Progress persistence.	

Technical/Performance Requirements
The application shall follow the Model–View–Controller (MVC) architectural pattern.
The application shall be built using HTML5, CSS3, and JavaScript as core front-end technologies.
The system shall use structured data storage (e.g., JSON data model or lightweight database) to manage animal information and discovery status.
The application shall be responsive and functional on: • Desktop screens (1024px and above) • Tablet screens (768px and above)
The application shall be compatible with modern web browsers
All source code shall be maintained in a GitHub repository organized according to MVC principles.
Task tracking shall be managed using GitHub Projects (Kanban board).